

Answers - Turning problems into quadratics (page ??)

1. $(2^2)^x + 2^x - 24 = 0$

$$(2^x)^2 + 2^x - 24 = 0$$

Making the substitution $u = 2^x$

$$u^2 + 2u - 24 = 0$$

$$u = 4.42, -5.42$$

2^x can never be negative so we can ignore the -5.42 solution.

$$2^x = 4.42$$

$$\ln 2^x = \ln 4.42$$

$$x \ln 2 = \ln 4.42$$

$$x = \frac{\ln 4.42}{\ln 2}$$

$$x = 2.14 \text{ (1dp)}$$

2. Rearrange to $9^x - 6^x - 4^x = 0$

We need a constant so divide through by the lowest term.

$$\frac{9^x}{4^x} - \frac{6^x}{4^x} - \frac{4^x}{4^x} = 0$$

$$\left(\frac{9}{4}\right)^x - \left(\frac{6}{4}\right)^x - 1 = 0$$

$$\left(\left(\frac{3}{2}\right)^2\right)^x - \left(\frac{3}{2}\right)^x - 1 = 0$$

$$\left(\left(\frac{3}{2}\right)^x\right)^2 - \left(\frac{3}{2}\right)^x - 1 = 0$$

Use the substitution $u = \left(\frac{3}{2}\right)^x$

$$u^2 - u - 1 = 0$$

$$u = 1.618, -0.618$$

$\left(\frac{3}{2}\right)^x$ can never be negative so we ignore -0.618.

$$\left(\frac{3}{2}\right)^x = 1.618$$

$$\ln \left(\frac{3}{2}\right)^x = \ln 1.618$$

$$x \ln \left(\frac{3}{2}\right) = \ln 1.618$$

$$x = \frac{\ln 1.618}{\ln \frac{3}{2}}$$

$$x = 1.187$$

3. We need a constant so divide through by the lowest term.

$$8\left(\frac{9^x}{4^x}\right) + 3\left(\frac{6^x}{4^x}\right) - 81 = 0$$

$$8\left(\frac{9}{4}\right)^x + 3\left(\frac{6}{4}\right)^x - 81 = 0$$

$$8\left(\left(\frac{3}{2}\right)^2\right)^x + 3\left(\frac{3}{2}\right)^x - 81 = 0$$

$$8\left(\left(\frac{3}{2}\right)^x\right)^2 + 3\left(\frac{3}{2}\right)^x - 81 = 0$$

Use the substitution $u = \left(\frac{3}{2}\right)^x$

$$8u^2 + 3u - 81 = 0$$

$$u = 3, -3.375$$

Since $\left(\frac{3}{2}\right)^x$ can never be negative, we can ignore the -3.375 solution.

$$\left(\frac{3}{2}\right)^x = 3$$

$$\ln\left(\frac{3}{2}\right)^x = \ln 3$$

$$x \ln\left(\frac{3}{2}\right) = \ln 3$$

$$x = \frac{\ln 3}{\ln \frac{3}{2}} = 2.71$$

4. We need a constant so divide through by the lowest term.

$$\left(\frac{25^x}{9^x}\right) + 2\left(\frac{15^x}{9^x}\right) - 24 = 0$$

$$\left(\frac{25}{9}\right)^x + 2\left(\frac{15}{9}\right)^x - 24 = 0$$

$$\left(\left(\frac{5}{3}\right)^2\right)^x + 2\left(\frac{5}{3}\right)^x - 24 = 0$$

$$\left(\left(\frac{5}{3}\right)^x\right)^2 + 2\left(\frac{5}{3}\right)^x - 24 = 0$$

Use the substitution $u = \left(\frac{5}{3}\right)^x$

$$u^2 + 2u - 24 = 0$$

$$u = 4, -6$$

Since $\left(\frac{5}{3}\right)^x$ can never be negative, we can ignore the -6 solution.

$$\left(\frac{5}{3}\right)^x = 4$$

$$\ln\left(\frac{5}{3}\right)^x = \ln 4$$

$$x \ln\left(\frac{5}{3}\right) = \ln 4$$

$$x = \frac{\ln 4}{\ln \frac{5}{3}} = 2.714$$

5. Multiply through by e^x :

$$e^{2x} - 20 = e^x$$

$$(e^x)^2 - 20 = 20e^x$$

$$(e^x)^2 + e^x - 20 = 0$$

$$e^x = -5, 4$$

Can't be negative, therefore:

$$e^x = 4$$

$$x = \ln 4$$

6. $3^{2x+1} - 82 \times 3^x + 27 = 0$

$$3^1 \times 3^{2x} - 82 \times 3^x + 27 = 0$$

$$3^1 \times (3^x)^2 - 82 \times 3^x + 27 = 0$$

$$u = 3^x$$

$$3u^2 - 82u + 27 = 0$$

$$u = \frac{1}{3}, 27$$

$$3^x = \frac{1}{3} \rightarrow x = -1$$

$$3^x = 27 \rightarrow x = 3x = -1, 3$$

7. $x^{\frac{1}{4}} + 2x^{-\frac{1}{4}} = 3$

$$(x^{\frac{1}{4}})^2 + 2 = 3x^{\frac{1}{4}}$$

$$(x^{\frac{1}{4}})^2 - 3x^{\frac{1}{4}} + 2 = 0$$

$$u = x^{\frac{1}{4}}$$

$$u^2 - 3u + 2 = 0$$

$$u = 1, 2$$

$$x^{\frac{1}{4}} = 1 \rightarrow x = 1$$

$$x^{\frac{1}{4}} = 2 \rightarrow x = 2^4 = 16$$

$$x = 1, 16$$

8. $x^2 - 5x + 6 + \frac{5}{x} + \frac{1}{x^2} = 0$ Group the terms with same powers: $x^2 + \frac{1}{x^2} - 5x + \frac{5}{x} + 6 = 0$

$$x^2 + \frac{1}{x^2} - 5(x + \frac{1}{x}) + 6 = 0 \text{ Use the substitution } u = x + \frac{1}{x} \quad (x + \frac{1}{x})^2 - 2 - 5(x + \frac{1}{x}) + 6 = 0$$

$$(x + \frac{1}{x})^2 - 5(x + \frac{1}{x}) + 4 = 0$$

$$u^2 - 5u + 4 = 0$$

$u = 1, 4$ Reverse substitution and solve:

$$x + \frac{1}{x} = 1$$

$$x^2 + 1 = x$$

$$x^2 - x + 1 = 0$$

$$x = \frac{1 \pm i\sqrt{3}}{2}$$

$$x + \frac{1}{x} = 4$$

$$x^2 + 1 = 4x$$

$$x^2 - 4x + 1 = 0$$

$$x = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3} \text{ Therefore, } x = \frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}, 2 + \sqrt{3}, 2 - \sqrt{3}$$

9. Set subject as $\sqrt{x+2}$ $x + 2 - 10\sqrt{x+2} + 22 = 0$

$$(\sqrt{x+2})^2 - 10\sqrt{x+2} + 22$$

$$u^2 - 10u + 22$$

$$u = \frac{-10 \pm \sqrt{12}}{2} = -5 \pm 2\sqrt{3}$$

$$\sqrt{x+2} = -5 \pm \sqrt{3} \text{ Solve for each separately: } \sqrt{x+2} = -5 + \sqrt{3}$$

$$x + 2 = 28 - 10\sqrt{3}$$

$$x = 26 - 10\sqrt{3}$$

$$\sqrt{x+2} = -5 - \sqrt{3}$$

$$x + 2 = 28 + 10\sqrt{3}$$

$$x = 26 + 10\sqrt{3}$$

$$\text{So the solutions are } x = 26 + 10\sqrt{3}, 26 - 10\sqrt{3}$$

10. Divide through by x^2 : $x^2 - 4x - 10 - \frac{4}{x} + \frac{1}{x^2} = 0$ Gather the terms with same powers:

$$(x^2 + \frac{1}{x^2}) - 4(x + \frac{1}{x}) - 10 = 0 \text{ Rewrite with } x + \frac{1}{x} \text{ as the subject: } (x + \frac{1}{x})^2 - 2 - 4(x + \frac{1}{x}) - 10 = 0$$

$$(x + \frac{1}{x})^2 - 4(x + \frac{1}{x}) - 12 = 0$$

$$u = x + \frac{1}{x}$$

$$u^2 - 4u - 12 = 0$$

$$u = -2, 6 \text{ Reverse substitution and solve: } x + \frac{1}{x} = -2$$

$$x^2 + 1 = -2x$$

$$x^2 + 2x + 1 = 0$$

$$x = -1$$

$$x + \frac{1}{x} = 6$$

$$x^2 + 1 = 6x$$

$$x^2 - 6x + 1 = 0$$

$$x = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$